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Overview

IGOR¹, b.next , and FindWhatMatters 

¹Find What Matters

Abstract

Keywords AI

Note

This article was written by a **human author**.

1. OVERVIEW

This report documents iterative AI-guided optimization by IGOR of the base PURE system to meet deliverable D1 of the AI-driven Cell-Free Energy Development and Optimization AI scientist project. The report is shared as a Developer Note, encapsulating human-readable information, digital data, and computational analysis. This page summarizes the key experimental data, model predictions, optimized PURE specification, and the project workflow. In addition, the Developer Note contains:

- Raw experimental data and recipes:
 - Data used and generated through iterative rounds is in the `data/` directory of the DevNote.
 - A full curated dataset is available as a CSV at [20260417-iteration-2-data_jl-curated.csv](#)
- Mechanistic interpretation of each round of experiments by IGOR:
 - [IGOR Report: Iteration 1 \(also independently here\)](#)
 - [IGOR Report: Iteration 2 \(also independently here\)](#)

2. WORKFLOW

Experimental data was collected over three rounds:

- First, we curated historical data from human-run experiments on base PURE system and PPK energy module implementation, to pre-seed the IGOR bayesian engine.

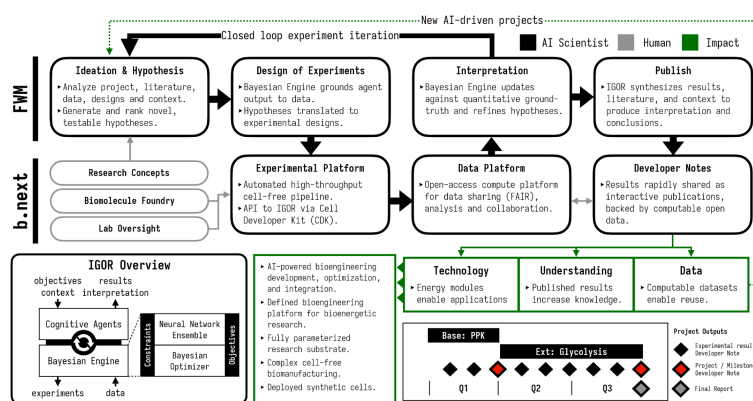


Figure 1. Overview of AI scientist optimization workflow. IGOR designs experiments which are run on the b.next experimental platform. Data from the platform is used to update the IGOR bayesian engine and informs further experiments, and also feeds the orchestrated research component for mechanistic interpretation. Intermediate results are published as Developer Notes.

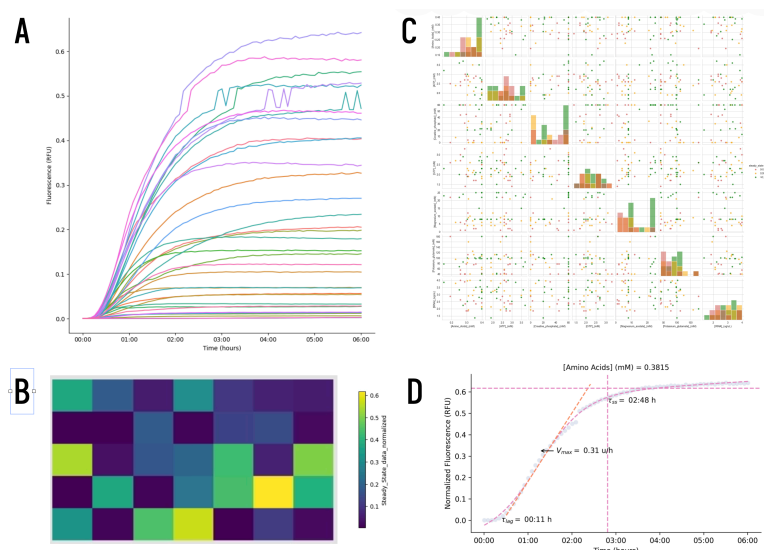


Figure 2. Normalized fluorescence data from Round 1 Discovery Plate. **A** Raw time-series fluorescence traces, one trace per discovery plate well. **B** Wellplate heatmap of steady-state fluorescence, determined by CDK kinetics analysis. **C** Pairwise analysis of discovery plate data, binned by steady-state fluorescence. **D** Kinetic analysis of top-performing composition experiment.

- Next, we operated a prototype experiment using a combination of IGOR-informed and manually-created PURE compositions and executed using a combination of automation and hand-running of experiments.
- Finally, we performed a full discovery plate round, designed by IGOR and collected using the automated experiment platform.

These rounds allowed us to prototype, test, and validate our experimental platform, the interfaces between b.next and FWM, and the ability of IGOR to interpret PURE compositional data.

3. EXPERIMENTAL DATA

The final Discovery Plate round examined 35 different compositions across 7 PURE parameters, producing yields from 0 to 0.62 normalized fluorescence units. The best-performing single composition was I14, with steady-state yield of 0.62 and maximum velocity of 0.31 (Figure 2).

4. GES (GENERATED EXPERIMENTAL SUGGESTIONS)

Experimental data produced in the most recent round was used to generate further predictions with IGOR. Including the PPK energy module, IGOR predicts a steady-state yield of 2.55 RFU; excluding the PPK energy module, it predicts a steady-state yield of 0.86 RFU.

5. PURE SPECIFICATIONS

Experiment (-PPK)

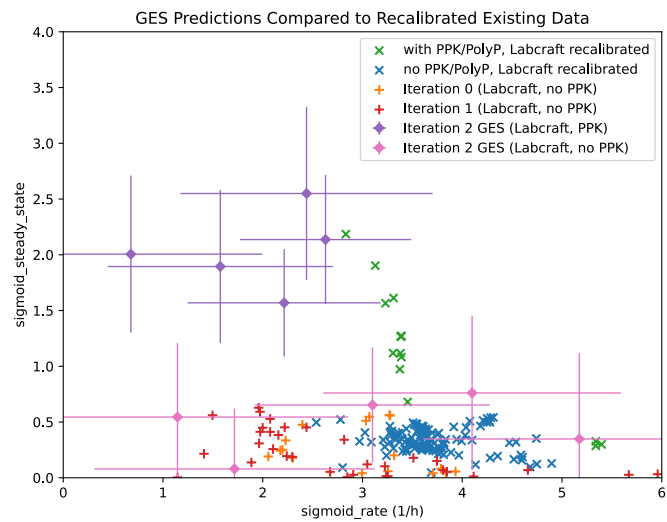


Figure 3. Generated Experimental Suggestions (GESes) predicted by IGOR for subsequent rounds of PURE optimization. The IGOR model was used to adjust data visualization to compensate for differences in experiment assembly and measurement.

Table 1. Highest-performing PURE composition (direct experiment)

Property	Value
ID	20260319-DiscoveryPlate-R2-I14
Well	I14
Date	3/19/2026 0:00
Experiment	20260319-DiscoveryPlate-R2
Name	DiscoveryPlate-R2-GES10
Type	Sample
PMix ID	AR-975
NEB in Pmix	FALSE
[PMix] (mg/mL)	1.8
Ribosome ID	AR-905
[Ribosome] (uM)	1.8
SMS ID	220260319-DiscoveryPlate-R2-SMS
tRNA ID	AR-902
[tRNA] (ug/uL)	2.8
DNA ID	AR-1028
[DNA] (ng/uL)	0
PMix Vol (uL)	0
Rxn Volume (uL)	10
[DNA] (nM)	5
[Magnesium acetate] (mM)	15
[Creatine phosphate] (mM)	53
[Potassium glutamate] (mM)	100
[ATP] (mM)	3.5
[GTP] (mM)	2.7
[ATP] (mM)	2.7
[GTP] (mM)	2.7

Predicted (-PPK)**Table 2.** *Highest-performing PURE composition (predicted, PPK excluded)*

Property	Value
Experiment	20260319-DiscoveryPlate-R2
NEB in Pmix	False
[PMix] (mg/mL)	1.80
[Ribosome] (uM)	3.60
[tRNA] (ug/uL)	4.46
[DNA] (nM)	9.40
[Magnesium acetate] (mM)	5.00
[Creatine phosphate] (mM)	54.33
[Potassium glutamate] (mM)	98.54
[ATP] (mM)	1.20
[GTP] (mM)	0.92
[Amino acid mix] (mM)	0.18
[PPK] (uM)	−0.00
[PolyP] (mM)	−0.00
[RNase Inhib] (U/mL)	2000.00
Reader Type	Cytation5
Gain	ext
Read Type	M
[PEG] (%)	−0.00
Product	deGFP
Condition	Labcraft
success prediction	0.39
sigmoid_steady_state prediction	0.86
sigmoid_rate (1/h) prediction	5.03
drift_rate (1/h) prediction	0.01
success uncertainty	0.39
sigmoid_steady_state uncertainty	0.46
sigmoid_rate (1/h) uncertainty	1.15
drift_rate (1/h) uncertainty	0.01

Predicted (+PPK)

Table 3. *Highest-performing PURE composition (predicted, PPK included)*

Property	Value
Experiment	20260319-DiscoveryPlate-R2
NEB in Pmix	False
[PMix] (mg/mL)	1.75
[Ribosome] (uM)	2.81
[tRNA] (ug/uL)	4.42
[DNA] (nM)	5.54
[Magnesium acetate] (mM)	5.00
[Creatine phosphate] (mM)	93.68
[Potassium glutamate] (mM)	97.19
[ATP] (mM)	1.67
[GTP] (mM)	1.17
[Amino acid mix] (mM)	0.26
[PPK] (uM)	2.52
[PolyP] (mM)	37.50
[RNase Inhib] (U/mL)	2000.00
Reader Type	Cytation5
Gain	ext
Read Type	M
[PEG] (%)	-0.00
Product	deGFP
Condition	Labcraft
success prediction	0.08
sigmoid_steady_state prediction	2.55
sigmoid_rate (1/h) prediction	2.21
drift_rate (1/h) prediction	0.06
success uncertainty	0.18
sigmoid_steady_state uncertainty	0.75
sigmoid_rate (1/h) uncertainty	1.27
drift_rate (1/h) uncertainty	0.02

6. DISCUSSION

We had several goals for the base module component of the project (months 1–3):

- Increase protein yield of the PURE system by optimizing its composition across its 108 protein, ribonucleic, and small molecule components.
- Prototype, test, and improve IGOR as an AI scientist, the automated experimental platform, and the interfaces between the two.
- Begin prototyping Developer Notes as a means of publishing human- and machine-readable reports on AI progress.
- Test and validate our iterative loop, to set up for further PURE optimization and energy module implementation in the extension (months 4–9) section of the project.

Through the first three months, we validated the ability of IGOR to design compositional PURE experiments, and the ability of the experimental platform to run them. Several teething problems limited our ability to run as many iterations as we initially expected: ss

Future upgrades:

- experimental platform constraints
- agentic output upgrades and automation of devnotes
- directly ingest raw data
- holdout of human v igor guesses